Further results on the brief *Physical vector spaces for quantum mechanics*

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Abstract

This note collects results that go beyond the technical brief *Physical vector spaces for quantum mechanics* [1] and partial answers to its open problems. They are not part of the brief. Each section is self-contained, and each names the Lean declarations that check it; the Lean files are in the brief's directory, under `QuantumVectorSpaces/FurtherResults/`.

1 The Hilbert brief's change of frame is not a map of Schwartz space

The brief proposes quantum vector spaces (QSTO:V) as a replacement for the Hilbert-space formulation QSTO:H, and argues in the proof of its Theorem 5 that QSTO:V satisfies [B]WDQ. The argument assumes that "the defining observables of A must be observables in B and vice-versa" for two frames A and B of the same system. This premise holds exactly when the two frames give the same topology on V (checked with the brief, `QuantumVectorSpaces.topology_eq_iff_continuous`). It is not automatic. This section shows that it fails for the brief's own Example 3, $V = \mathcal{S}(\mathbb{R})$, with the change of frame that the companion brief *Unphysicality of Hilbert spaces for quantum mechanics* [2] uses against QSTO:H.

That change of frame is $y = \tan(\frac{\pi}{2}\operatorname{erf} x)$. Its unitary U on $L^2(\mathbb{R})$ sends the Gaussian packet

$$\psi(x) = \left(e^{-x^2}/\sqrt{\pi}\right)^{1/2} = \pi^{-1/4}e^{-x^2/2} \quad \text{to} \quad \phi(y) = \left(\pi(1+y^2)\right)^{-1/2}.$$

Theorem 1.1. *1. ψ is a Schwartz function, and $U\psi = \phi$ is not. So U does not map $\mathcal{S}(\mathbb{R})$ into itself.*

2. The position Y of the second frame, carried back to the first as $U^{-1}YU$, is not defined at the Schwartz function ψ.

Proof. Every Schwartz function f lies in the domain of the position operator, since xf is again a Schwartz function and so in L^2. But $\int y^2|\phi(y)|^2\,dy = \int y^2/(\pi(1+y^2))\,dy = \infty$, so ϕ is not in that domain (the brief *Unphysicality of Hilbert spaces*, Proposition 3). This gives (1). For (2), $\psi \in D(U^{-1}YU)$ iff $U\psi \in D(Y)$, which fails. □

So the two observers do not describe the same V with two families of defining observables, as the premise requires: the second observer cannot express the first observer's ψ in $\mathcal{S}(\mathbb{R})$. QSTO:V gives [B]WDQ only for changes of frame whose unitaries map V onto itself. The restriction is stated in print by Carcassi, Calderón and Aidala: on the Schwartz space "the only unitary transformations we are interested in are those that map Schwartz functions to Schwartz functions", and the change

of coordinate "is automatically ruled out" [3, Appendix, Sec. VIII]. The brief does not repeat it. Theorem 1.1 is the concrete instance, checked in Lean.

Which changes of variable do map $\mathcal{S}(\mathbb{R})$ into itself is known: Galbis and Jordá characterize them [4, Thm. 2.3]. That the restriction cannot be lifted by choosing a different space is shown in the companion note on the Hilbert brief [5, Sec. 2]: no nonzero state of $L^2(\mathbb{R})$ has finite position variance in every smooth frame both now and after a free evolution. A quantum vector space has to come with its group of changes of frame.

1.1 Lean

Module:
`QuantumVectorSpaces.FurtherResults.SchwartzFrame`,
namespace `QuantumVectorSpaces.FurtherResults`. The unitary U (`frameUnitary`), the packets and the position operator are those of the Hilbert brief's formalization.

- `toLp_gaussSchwartz`: ψ is $\pi^{-1/4}$ times TauCeti's Gaussian Schwartz map.
- `toLp_mem_positionDomain`: every Schwartz function lies in the domain of the position operator on $L^2(\mathbb{R})$.
- `frameUnitary_gaussState_not_mem_range`, `not_forall_frameUnitary_mem_range`: Theorem 1.1(1).
- `toLp_gaussSchwartz_not_mem_domain`: Theorem 1.1(2).

The statement is made in $L^2(\mathbb{R})$, where U lives; it does not need U to be smooth.

References

[1] G. Carcassi, *Physical vector spaces for quantum mechanics*, technical brief, Assumptions of Physics, `https://assumptionsofphysics.org/resources/briefs`.

[2] G. Carcassi, *Unphysicality of Hilbert spaces for quantum mechanics*, technical brief, Assumptions of Physics, `https://assumptionsofphysics.org/resources/briefs`.

[3] G. Carcassi, F. Calderón, C. A. Aidala, The unphysicality of Hilbert spaces, Quantum Stud.: Math. Found. **12**, 13 (2025), arXiv:2308.06669v3.

[4] A. Galbis, E. Jordá, Composition operators on the Schwartz space, Rev. Mat. Iberoam. **34** (2018) 397–412.

[5] R. Stephan, *Further results on the brief "Unphysicality of Hilbert spaces for quantum mechanics"*, `briefs/HilbertUnphysicality/further-results.pdf`.